Tugas Praktikum Pemrograman Mobile
Pertemuan Ke - 1



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Universitas Esa Unggul

Praktikum Flutter

1. Tujuan

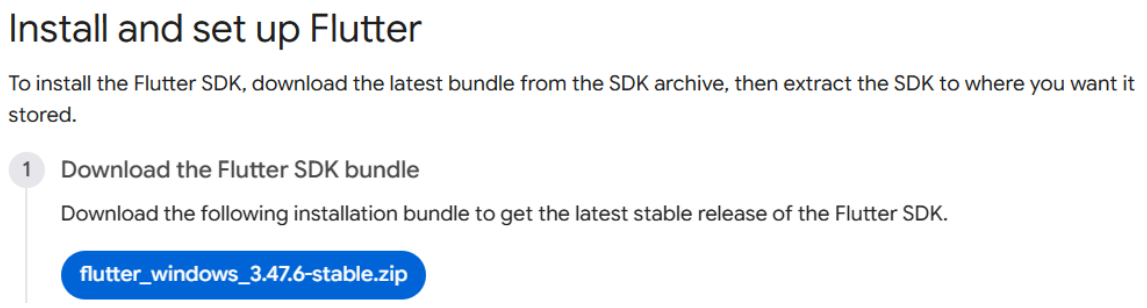
Praktikum Flutter pertama ini bertujuan agar kita dapat memahami dasar pengembangan aplikasi menggunakan Flutter. Kita diperkenalkan dengan Flutter sebagai UI toolkit dan Dart sebagai bahasa pemrogramannya, kemudian belajar menyiapkan Flutter SDK, Android Studio, serta emulator / perangkat Android. Setelah environment siap, kita akan membuat project Flutter pertama, menjalankan aplikasi, memahami struktur folder project, dan mempelajari konsep widget sebagai dasar pembentukan tampilan aplikasi.

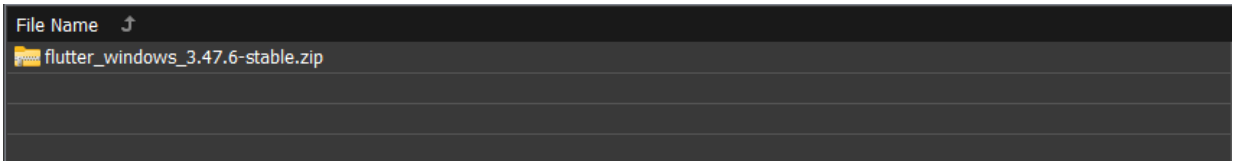
2. Alat dan Bahan

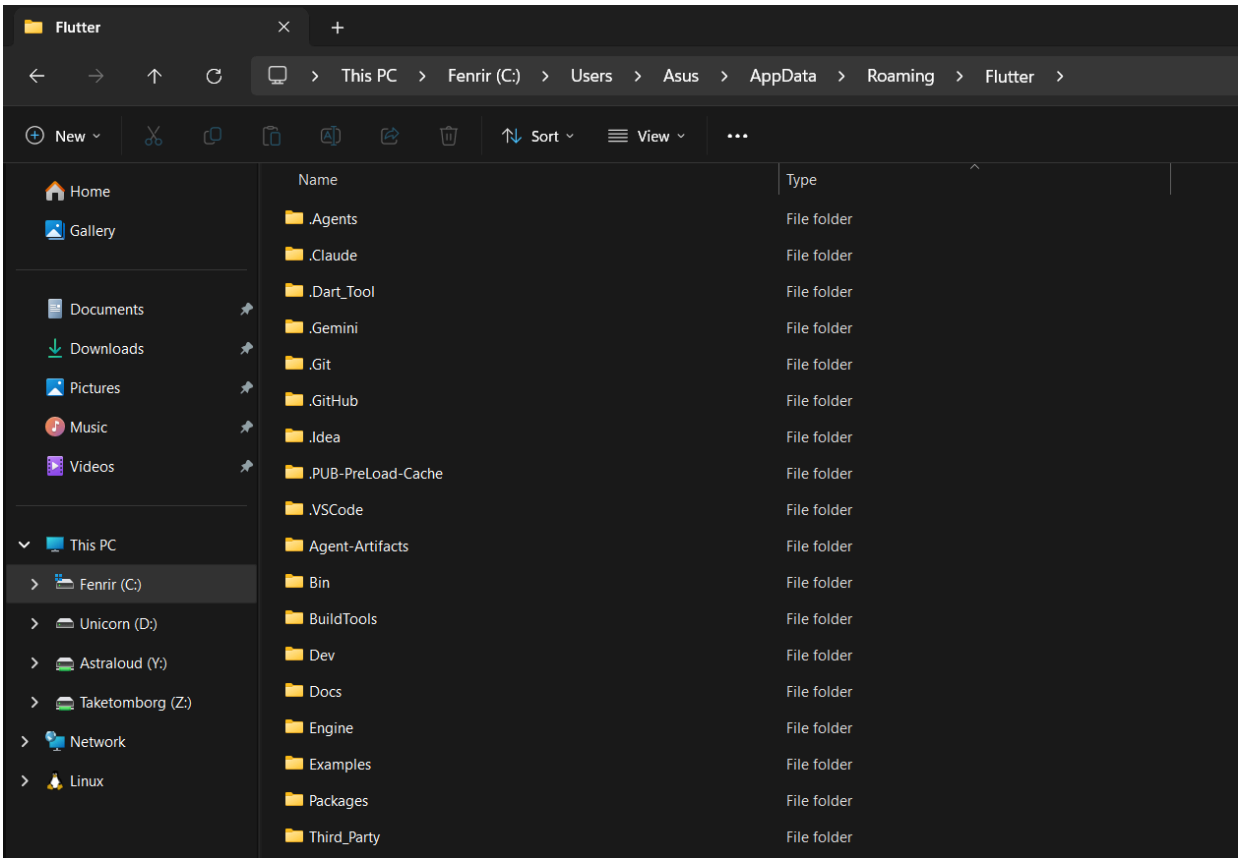
- Komputer Atau Laptop
- Flutter Package (Flutter - Install)
- Android Studio / IDE Dengan Support Flutter & Dart
- Emulator SDK Android / HP Android Dengan Debugging Mode

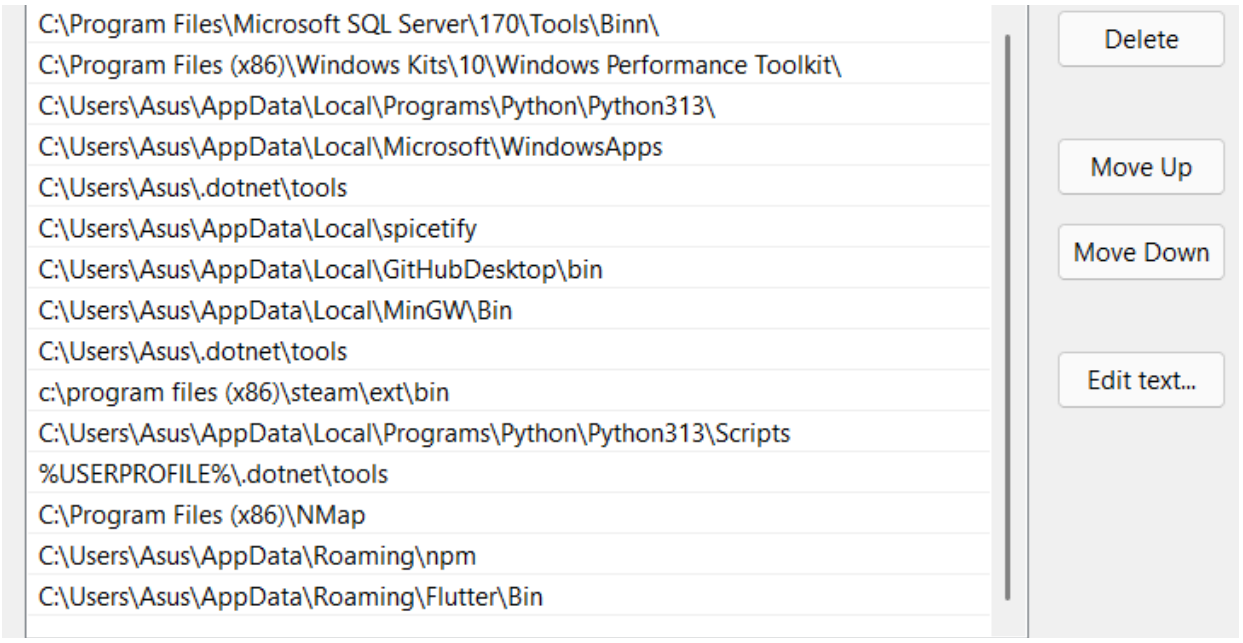
3. Tahapan Instalasi Awal dan Setup

3.1 Instalasi Flutter Di Windows dan Penambahan Ke Path :

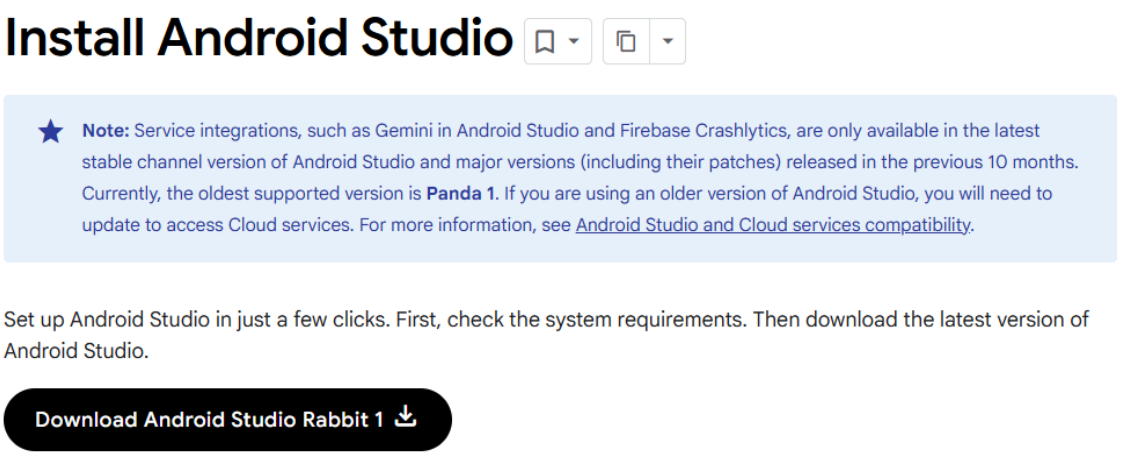


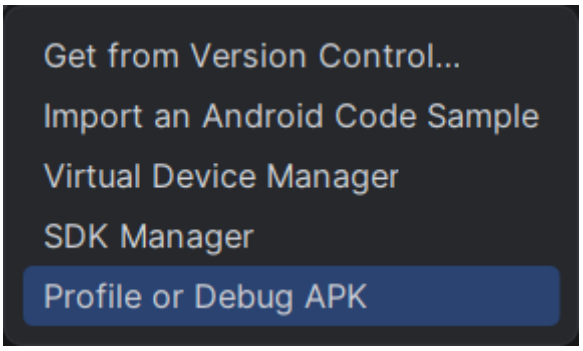

File Name
flutter_windows_3.47.6-stable.zip

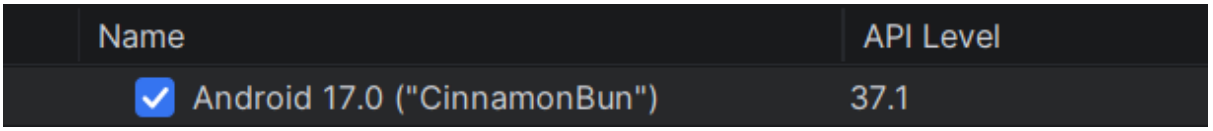

Flutter
This PC
Fenrir (C:)
Users
Asus
AppData
Roaming
Flutter
New
Sort
View
Home
Gallery
Documents
Downloads
Pictures
Music
Videos
This PC
Fenrir (C:)
Unicorn (D:)
Astraloud (Y:)
Taketomborg (Z:)
Network
Linux
Name
Type
.Agents
.Claude
.Dart_Tool
.Gemini
.Git
.GitHub
.Idea
.PUB-PreLoad-Cache
.VSCode
Agent-Artifacts
Bin
BuildTools
Dev
Docs
Engine
Examples
Packages
Third_Party
File folder

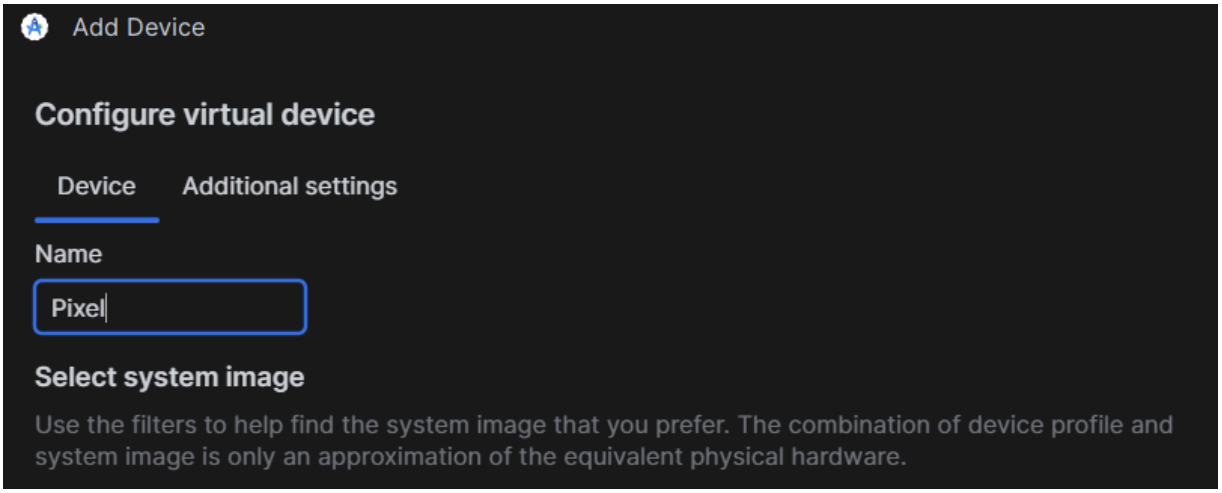

C:\Program Files\Microsoft SQL Server\170\Tools\Binn\
C:\Program Files (x86)\Windows Kits\10\Windows Performance Toolkit\
C:\Users\Asus\AppData\Local\Programs\Python\Python313\
C:\Users\Asus\AppData\Local\Microsoft\WindowsApps
C:\Users\Asus\.dotnet\tools
C:\Users\Asus\AppData\Local\spicetify
C:\Users\Asus\AppData\Local\GitHubDesktop\bin
C:\Users\Asus\AppData\Local\MinGW\Bin
C:\Users\Asus\.dotnet\tools
c:\program files (x86)\steam\ext\bin
C:\Users\Asus\AppData\Local\Programs\Python\Python313\Scripts
%USERPROFILE%\.dotnet\tools
C:\Program Files (x86)\NMap
C:\Users\Asus\AppData\Roaming\npm
C:\Users\Asus\AppData\Roaming\Flutter\Bin
Delete
Move Up
Move Down
Edit text...

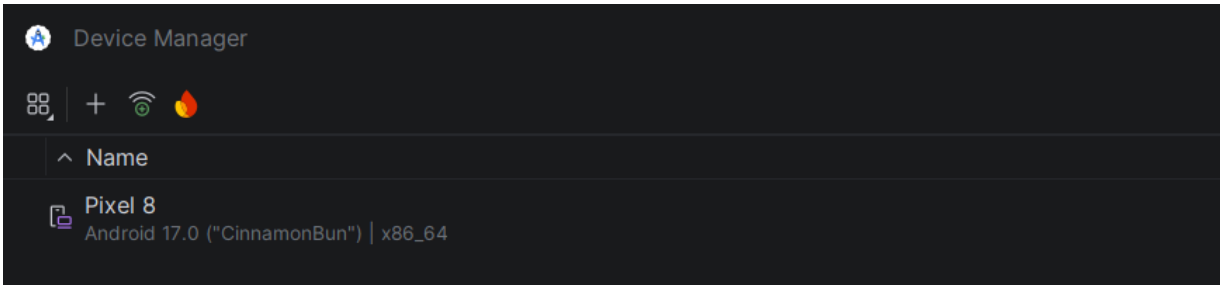
3.2 Instalasi Android Studio Dengan Device Emulator :



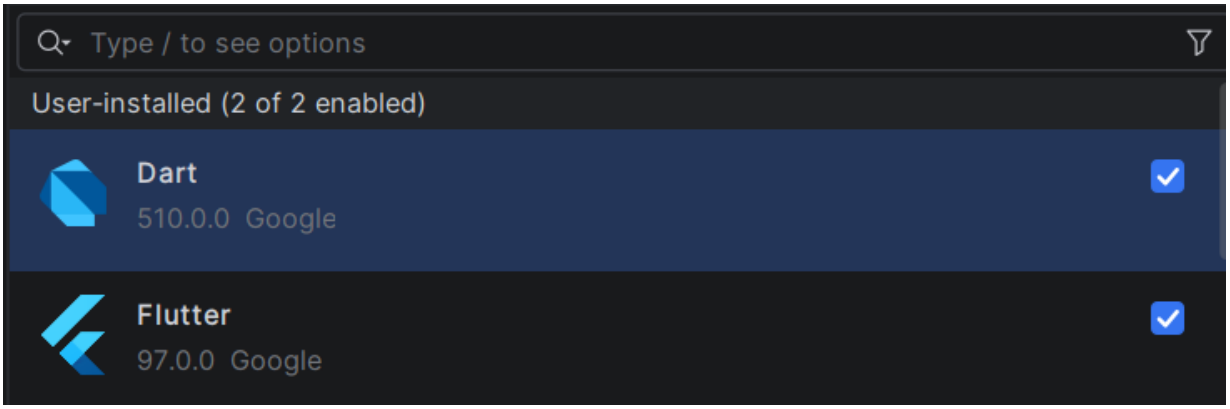








3.3 Pemasangan Ekstensi Flutter dan Dart di Android Studio :



3.4 Pengecekan Status Flutter di Command Prompt :

```
C:\Windows\System32>flutter --version
Flutter 3.47.6 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 5fc346839b (2 days ago) • 2026-09-30 15:02:49 -0700
Engine • hash b8c8d3d8d5d0095127057f8a29ca8cc53da2167c (revision 692136cb65) (2 days ago) • 2026-09-30 00:56:59.000Z
Tools • Dart 3.13.5 • DevTools 2.60.0
```

```
C:\Windows\System32>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[√] Flutter (Channel stable, 3.47.6, on Microsoft Windows [Version 10.0.26200.9550], locale en-US)
[√] Windows Version (Windows 11 or higher, 25H2, 2009)
[√] Android toolchain - develop for Android devices (Android SDK version 37.0.0)
[X] Chrome - develop for the web (Cannot find Chrome executable at .\Google\Chrome\Application\chrome.exe)
    ! Cannot find Chrome. Try setting CHROME_EXECUTABLE to a Chrome executable.
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its default components
[√] Connected device (2 available)
[√] Network resources
```

4. Pengenalan Flutter dan Device Emulator :

4.1 Pembuatan Folder Proyek dan Pengenalan Struktur :

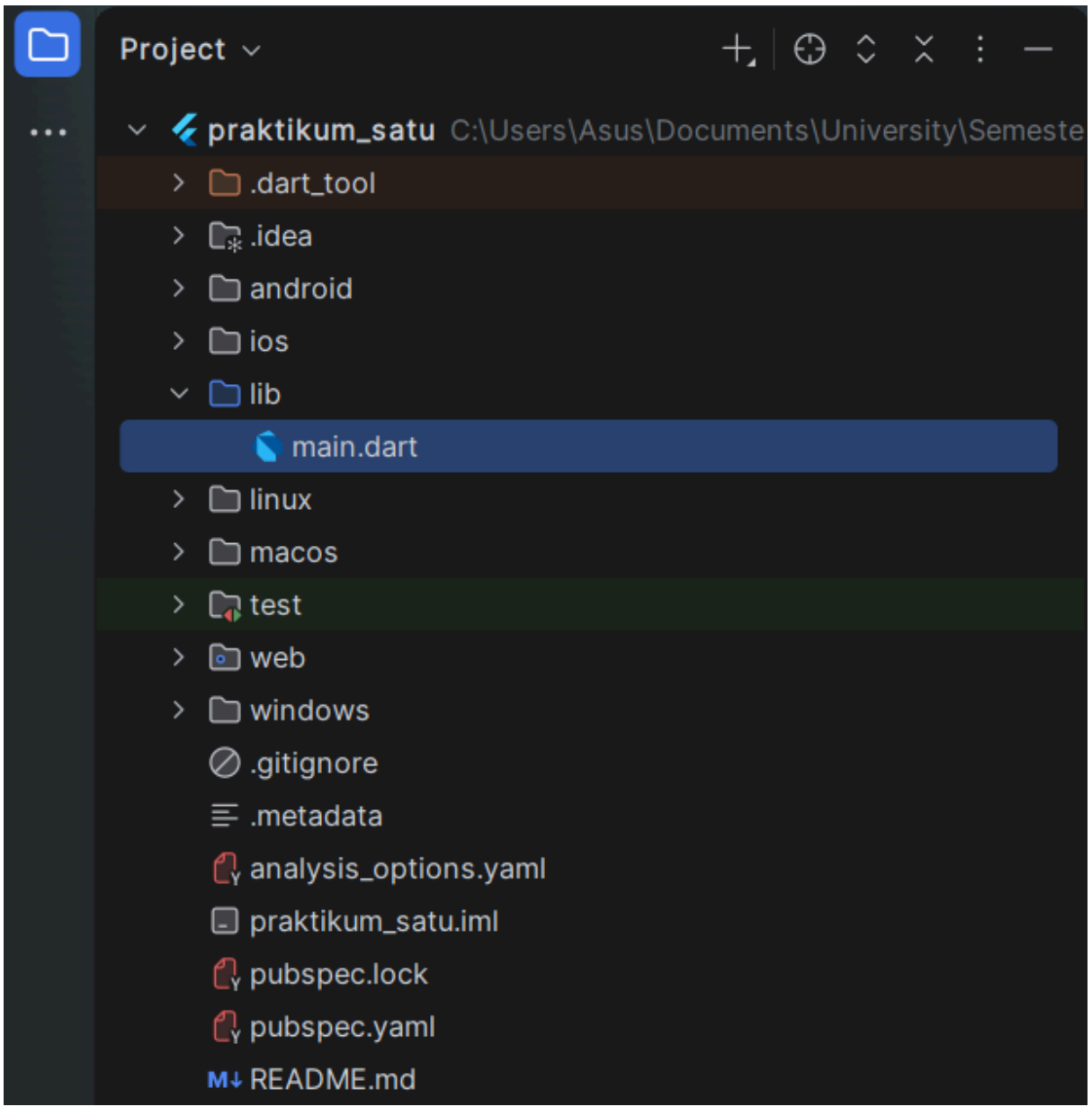
```
C:\Users\Asus\Documents\University\Semester 5\Pemrograman Mobile\Praktikum>flutter create praktikum_satu
Creating project praktikum_satu...
Resolving dependencies in `praktikum_satu`...
Downloading packages...
Got dependencies in `praktikum_satu`.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

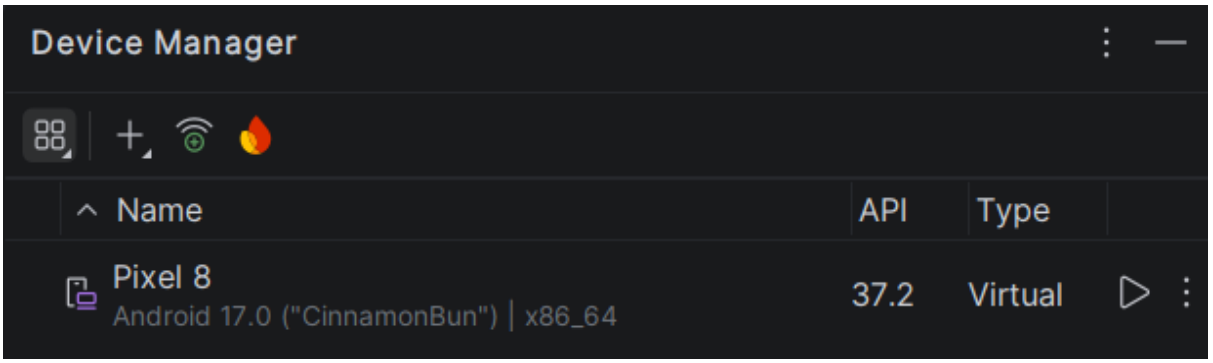
In order to run your application, type:

  $ cd praktikum_satu
  $ flutter run

Your application code is in praktikum_satu\lib\main.dart.
```



4.2 Aktivasi Emulator Untuk Tampilan Default / Template :

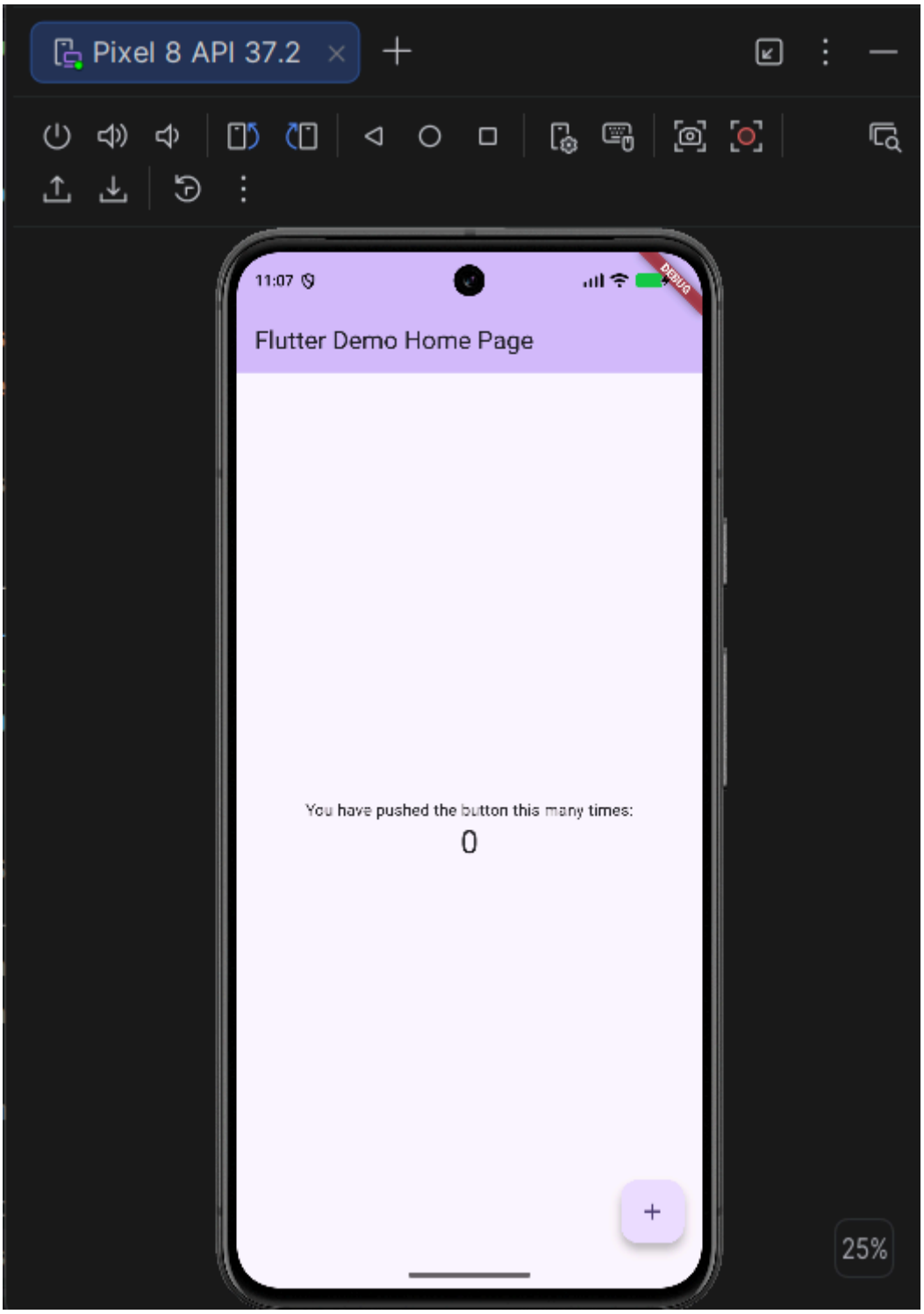




```
Terminal  Local

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Asus\Documents\University\Semester 5\Pemrograman Mobile\Praktikum\praktikum_satu> flutter run -v
```



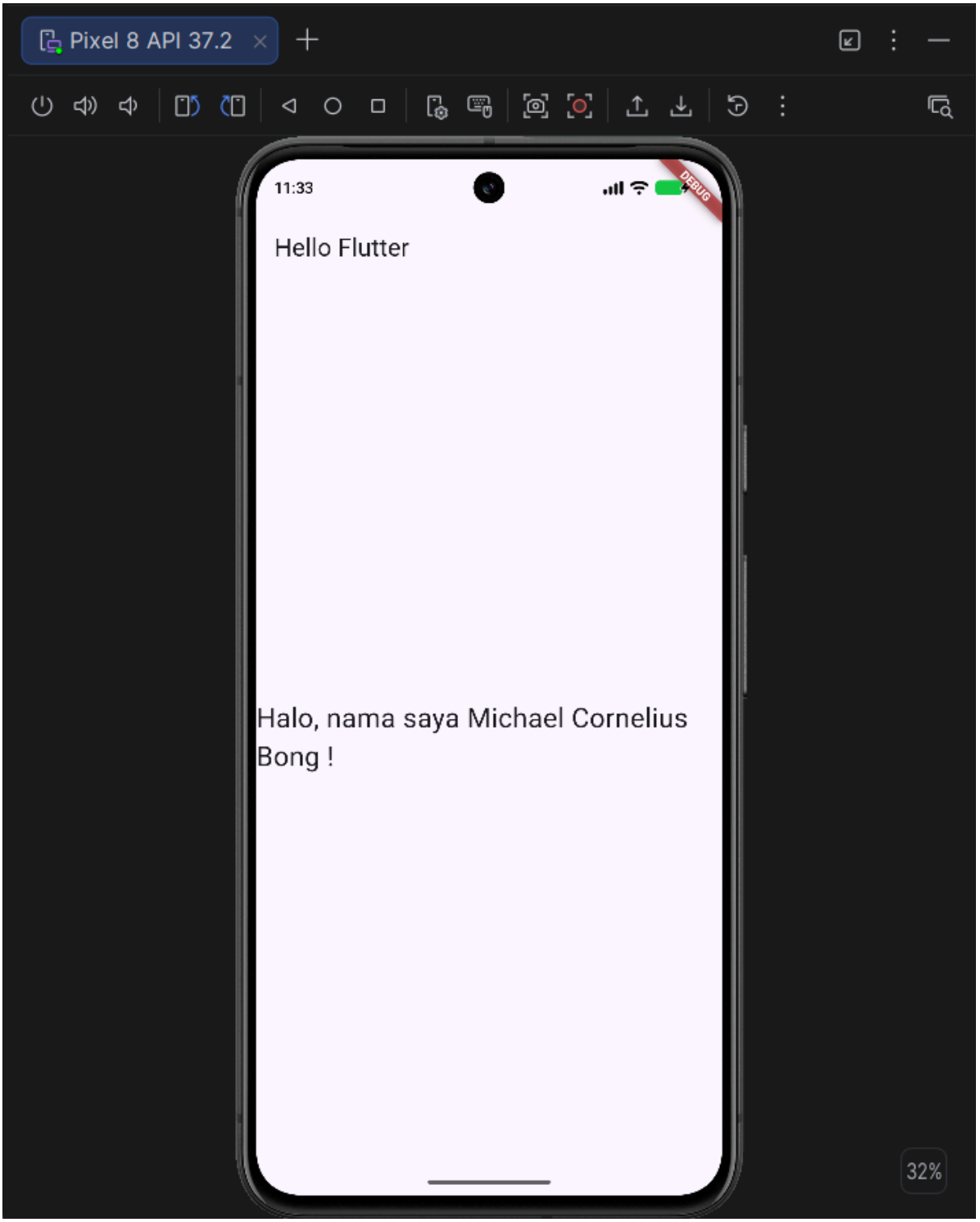
5. Pembuatan Aplikasi Dasar “Hello Flutter”

5.1 Pengubahan Pada Seluruh Isi Main Dart File :

main.dart

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: Scaffold(
        appBar: AppBar(title: const Text('Hello Flutter')),
        body: const Center(
          child: Text(
            'Halo, nama saya Michael Cornelius Bong !',
            style: TextStyle(fontSize: 24),
          ), Text
        ), Center
      ), Scaffold
    ); MaterialApp
  }
}
```

```
Performing hot reload...
Reloaded 0 libraries in 274ms (compile: 25 ms, reload: 0 ms, reassemble: 73 ms).
```

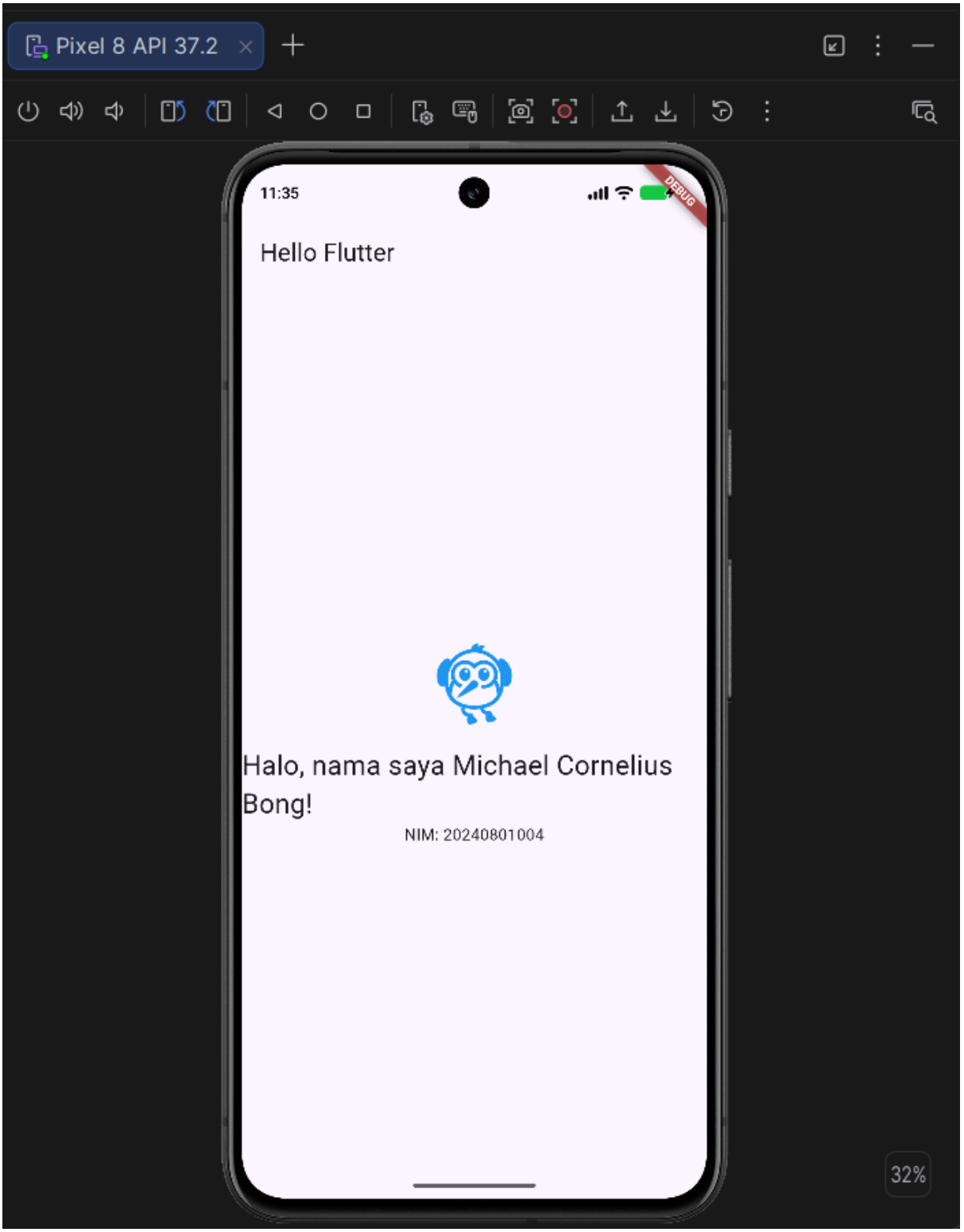

Pixel 8 API 37.2
11:33
DEBUG
Hello Flutter
Halo, nama saya Michael Cornelius Bong !
32%

5.2 Penambahan Code Pada Body :

main.dart

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}
class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: Scaffold(
        appBar: AppBar(title: const Text('Hello Flutter')),
        body: Center(
          child: Column(
            mainAxisAlignment: MainAxisAlignment.center,
            children: const [
              Icon(Icons.flutter_dash, size: 80, color: Colors.blue),
              SizedBox(height: 16),
              Text('Halo, nama saya Michael Cornelius Bong!', style: TextStyle(fontSize: 24)),
              Text('NIM: 20240801004'),
            ],
          ), Column
        ), Center
      ), Scaffold
    ); MaterialApp
  }
}
```

```
Performing hot reload...
Reloaded 1 of 754 libraries in 11,215ms (compile: 54 ms, reload: 219 ms, reassemble: 23 ms).
```


Pixel 8 API 37.2
11:35
DEBUG
Hello Flutter
Halo, nama saya Michael Cornelius Bong!
NIM: 20240801004
32%

5.3 Pembuatan Class CounterPage dan Menggunakannya :

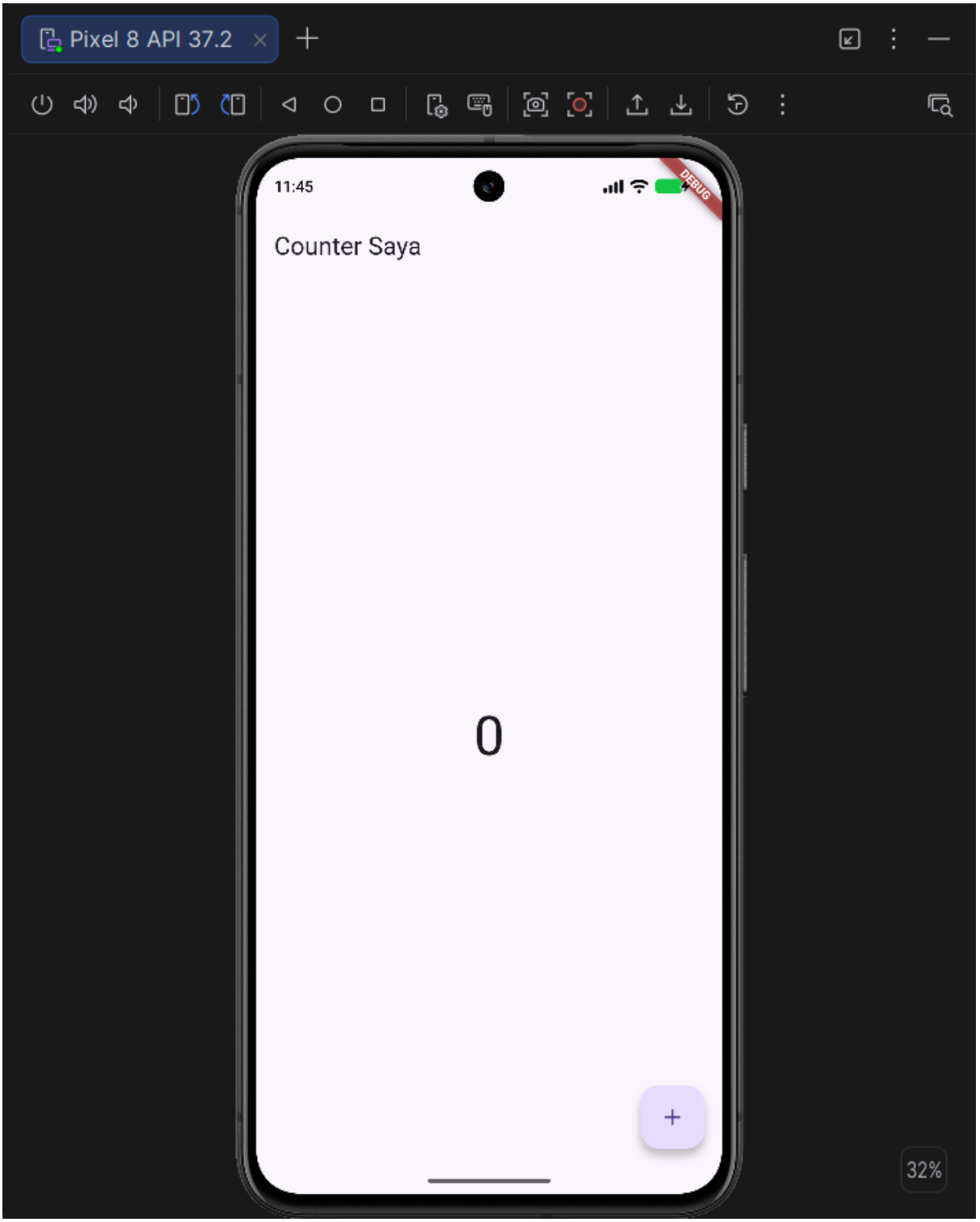
main.dart

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: const CounterPage(),
    ); MaterialApp
  }
}

class CounterPage extends StatefulWidget {
  const CounterPage({super.key});
  @override
  State<CounterPage> createState() => _CounterPageState();
}
class _CounterPageState extends State<CounterPage> {
  int _count = 0;
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Counter Saya')),
      body: Center(
        child: Text('$_count', style: const TextStyle(fontSize: 48)),
      ), Center
      floatingActionButton: FloatingActionButton(
        onPressed: () => setState(() => _count++),
        child: const Icon(Icons.add),
      ), FloatingActionButton
    ); Scaffold
  }
}
```

```
Performing hot reload...
Reloaded 1 of 754 libraries in 1,235ms (compile: 33 ms, reload: 399 ms, reassemble: 607 ms).
```


Pixel 8 API 37.2
11:45
DEBUG
Counter Saya
0
+
32%

6. Latihan Mandiri

1. Ubah warna AppBar dan warna teks sesuai selera (backgroundColor, color).
2. Tambahkan tombol kedua (Icons.remove) untuk mengurangi angka.
3. Tambahkan tombol reset yang mengembalikan angka ke 0.
4. Cegah angka menjadi negatif.

main.dart

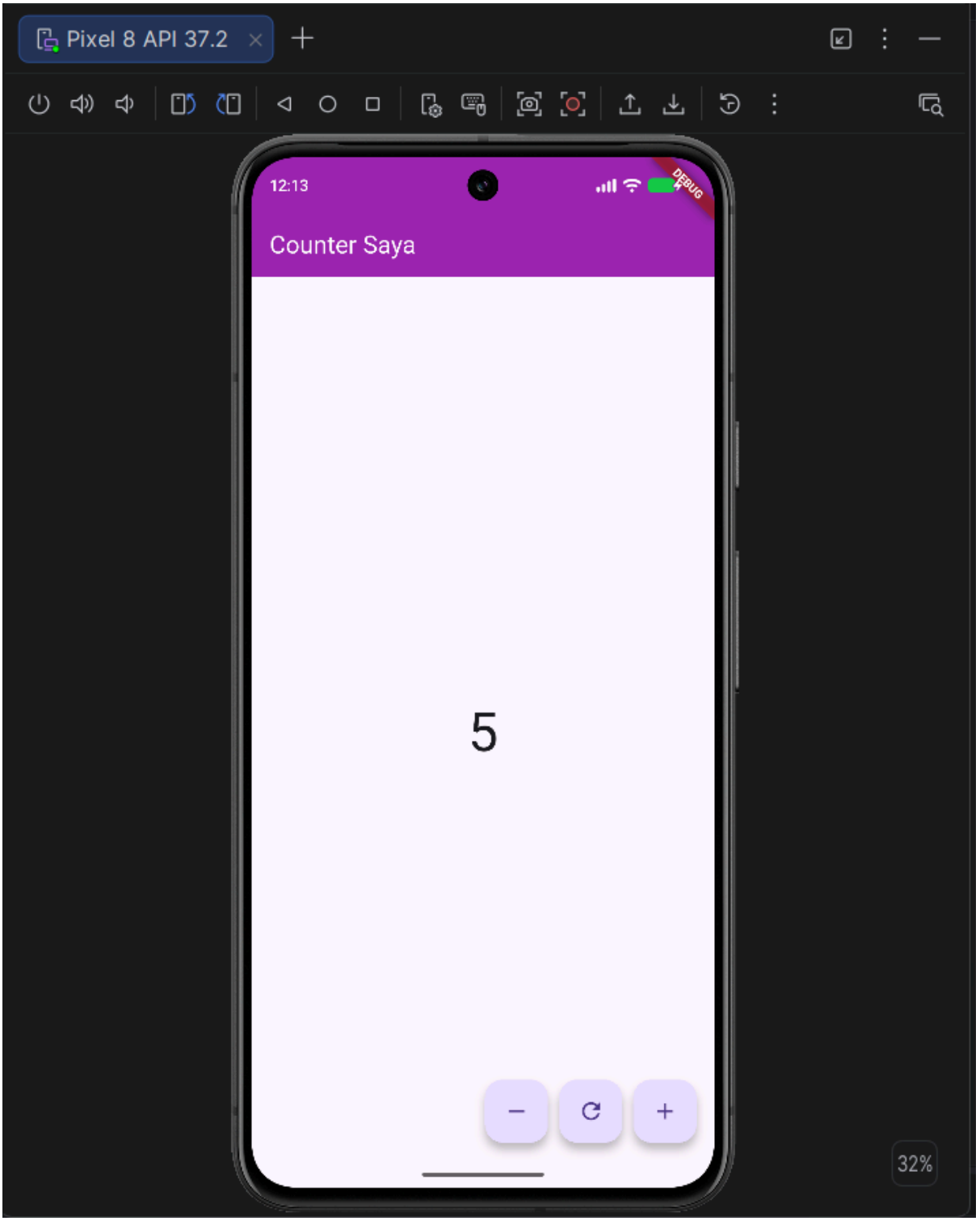
```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Praktikum 1',
      home: const CounterPage(),
    ); MaterialApp
  }
}

class CounterPage extends StatefulWidget {
  const CounterPage({super.key});
  @override
  State<CounterPage> createState() => _CounterPageState();
}
class _CounterPageState extends State<CounterPage> {
  int _count = 0;
  void _increment() {
    setState(() => _count++);
  }
  void _decrement() {
    setState(() {
      if (_count > 0) {
        _count--;
      }
    });
  }
  void _reset() {
    setState(() => _count = 0);
  }
```

```dart
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        backgroundColor: Colors.purple,
        title: const Text(
          'Counter Saya',
          style: TextStyle(color: Colors.white),
        ), Text
      ), AppBar
      body: Center(
        child: Text('$_count', style: const TextStyle(fontSize: 48)),
      ), Center
      floatingActionButton: Row(
        mainAxisAlignment: MainAxisAlignment.end,
        children: [
          FloatingActionButton(
            heroTag: 'minus',
            onPressed: _decrement,
            child: const Icon(Icons.remove),
          ), FloatingActionButton
          const SizedBox(width: 10),
          FloatingActionButton(
            heroTag: 'reset',
            onPressed: _reset,
            child: const Icon(Icons.refresh),
          ), FloatingActionButton
          const SizedBox(width: 10),
          FloatingActionButton(
            heroTag: 'plus',
            onPressed: _increment,
            child: const Icon(Icons.add),
          ), FloatingActionButton
        ],
      ), Row
    ); Scaffold
  }
}
```

```
Performing hot reload...
Reloaded 1 of 754 libraries in 907ms (compile: 39 ms, reload: 439 ms, reassemble: 220 ms).
```


Pixel 8 API 37.2
12:13
DEBUG
Counter Saya
5
32%

7. Tugas

Buat aplikasi "Kartu Perkenalan" satu halaman yang menampilkan: foto / ikon, nama, NIM, jurusan, dan hobi menggunakan Column, Text, Icon, dan SizedBox. Kumpulkan berupa tangkapan layar aplikasi berjalan dan tautan repositori (atau berkas main.dart).

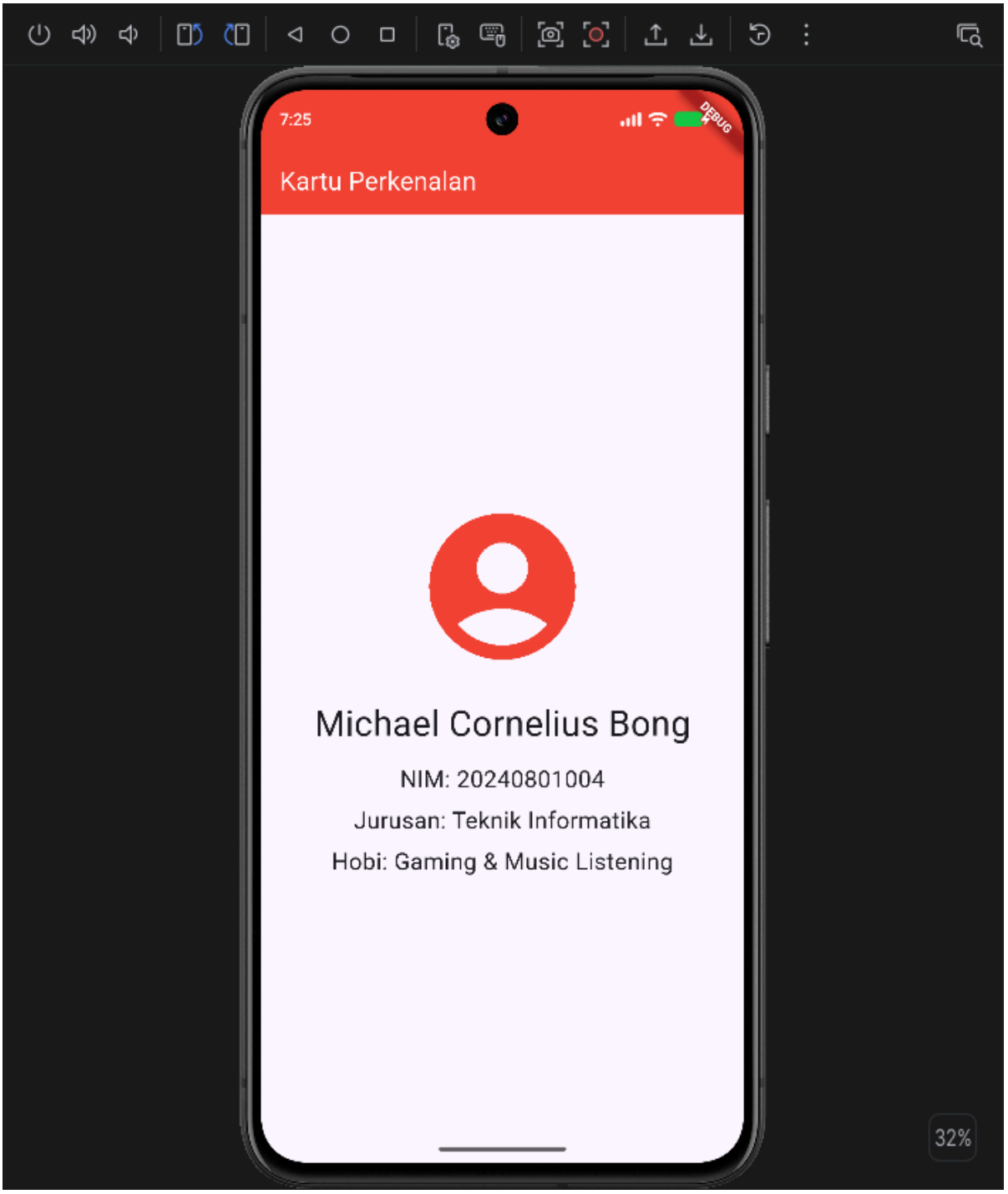
main.dart

```dart
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Kartu Perkenalan',
      home: const KartuPerkenalan(),
    ); MaterialApp
  }
}

class KartuPerkenalan extends StatelessWidget {
  const KartuPerkenalan({super.key});
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Kartu Perkenalan'),
        backgroundColor: Colors.indigo,
        foregroundColor: Colors.white,
      ), AppBar
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
            const Icon(Icons.account_circle, size: 120, color: Colors.indigo),
            const SizedBox(height: 20),
            const Text(
              'Michael Cornelius Bong',
              style: TextStyle(fontSize: 30),
            ), Text
            const SizedBox(height: 12),
            const Text(
              'NIM: 20240801004',
              style: TextStyle(fontSize: 18),
            ), Text
            const SizedBox(height: 6),
```

```
43                const Text(
44                  'Jurusan: Teknik Informatika',
45                  style: TextStyle(fontSize: 18),
46                ), Text
47                const SizedBox(height: 6),
48                const Text(
49                  'Hobi: Gaming & Music Listening',
50                  style: TextStyle(fontSize: 18),
51                ), Text
52              ],
53            ), Column
54          ), Center
55        ); Scaffold
56      }
57    }
```



8. Pertanyaan Refleksi

1. Apa perbedaan StatelessWidget dan StatefulWidget?

StatelessWidget digunakan untuk widget yang tampilannya tidak berubah atau statis. Sedangkan, StatefulWidget digunakan untuk widget yang tampilannya dapat berubah karena adanya perubahan kondisi atau situasi.

2. Mengapa perubahan variabel _count perlu dibungkus setState()?

Perubahan dibungkus dengan setState() agar Flutter mengetahui bahwa state / kondisi dari widget yang berhubungan dengan _count telah berubah dan tampilannya perlu diperbarui.

3. Apa keuntungan hot reload dibanding rebuild penuh?

Keuntungan dalam menggunakan hot reload adalah perubahan yang kita buat pada kode dapat langsung dilihat pada aplikasi tanpa harus melakukan restart aplikasi secara penuh atau melakukan flutter run

9. Catatan Pribadi

- Flutter : UI toolkit untuk membuat aplikasi dari satu basis kode yaitu Dart.
- Dart : Bahasa pemrograman yang digunakan oleh Flutter.
- Widget : Komponen yang digunakan untuk membangun tampilan aplikasi Flutter.
- MaterialApp : Widget utama yang menjadi sebuah layer dalam aplikasi Flutter.
- Scaffold : Struktur dasar sebuah halaman aplikasi.
- AppBar : Bagian atas halaman yang biasanya berisi judul aplikasi.
- Center : Widget untuk menempatkan isi di bagian tengah layar.
- Column : Widget untuk menyusun beberapa widget secara vertikal.
- Text : Widget untuk menampilkan teks pada aplikasi.
- Icon : Widget untuk menampilkan ikon.
- SizedBox : Widget untuk memberikan jarak atau ruang antar widget.
- StatelessWidget : Widget yang tampilannya tidak berubah atau bersifat statis.
- StatefulWidget : Widget yang tampilannya berubah karena perubahan kondisi / state.
- setState() : Untuk memberitahu bahwa state berubah dan tampilan perlu diperbarui.
- Hot Reload : Untuk melihat perubahan kode secara langsung tanpa melakukan restart.
- Widget Tree : Susunan widget yang membentuk tampilan aplikasi.
- Flutter Doctor : Perintah untuk memeriksa apakah environment Flutter sudah siap.
- Flutter Run : Perintah untuk menjalankan aplikasi Flutter pada device yang ada.